
Final Python Project

Water Quality Prediction

Group : WaterRangers

Project Title : Water Quality Prediction

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Dataset Kaggle :

Water Quality : The water_potability.csv file contains water quality metrics for 3276 different water bodies.

<https://www.kaggle.com/datasets/adityakadiwal/water-potability/>

Context :

In a world filled with inequalities, getting access to safe drinking-water is essential to health, a basic human right and a component of effective policy for health protection. This is important as a life and development issue at a national, regional and local level. For this reason, many institutions promote sustainability and ecological progress. In the lens of KMUTT's goals, the school has been committed to three core benchmarks regarding water, mainly focused on sustainability, recycling or consumption reduction.

In an attempt to diversify the research, we've chosen to address the sanitary aspect of water usage by analyzing its quality.

For more information, check the KMUTT Sustainable Water goals page :

<https://sustainable.kmutt.ac.th/goals6/>

Problematic :

Traditionally, to certify water as potable, treatment stations must send samples to a laboratory for bacterial and chemical analysis, which can take 24 to 48 hours. During this waiting period, there is a risk that contaminated water may be distributed.

Solution :

In order to reduce delays and save money in review of test results, we will train a machine learning model to predict whether water is safe to drink, based on its characteristics: pH, hardness, solids, chloramines, sulphate, conductivity, organic carbon, trihalomethanes, and turbidity. We won't waste any more time going back and forth to the lab.